Anshika Bajpai

United States | anshikabajpai23@gmail.com | Github | linkedin.com/in/anshikabajpai23/ | Portfolio | +1 (930) 204 3030

SUMMARY

Senior Software Engineer and **Machine Learning Engineer** with **3+ years** of experience building scalable AI systems and ML infrastructure. Experienced in LLMs, computer vision, and distributed ML pipelines. Proven track record deploying production systems supporting 10K+ users and optimizing large-scale enterprise platforms. Currently researching generative models and medical imaging AI at Indiana University.

EDUCATION

Indiana University Bloomington, Master of Science in Data Science, USA **May 2026**

Coursework: *Machine Learning, Algorithms, Statistics, Advance Database Concepts, Exploratory Data Analysis, Data Mining*

Bachelor of Technology in Computer Science and Engineering, Jaypee Institute of Information Technology, India **May 2021**

WORK EXPERIENCE

Machine learning research Assistant, Indiana University (AIMed Lab) **Jan 2026 – present**

- Led a medical imaging and computer vision research project using ***GANs*** and ***diffusion models*** to ***generate counterfactual healthy prostate MRIs***, enabling ***patient-specific digital twins*** for longitudinal treatment outcome evaluation.

Machine Learning Engineer Intern, Palo Alto Networks, California **May 2025 - Aug 2025**

- Developed a ***full-stack document ingestion portal***, integrating backend APIs and utilizing ***Retrieval-Augmented Generation (RAG)*** with Top-K ranking and caching for real-time, accurate information retrieval, supporting ***10,000+ Palo Alto Networks employees.***
- Designed and engineered ***prompt engineering pipelines and multi-turn conversational agents*** to enhance Copilot contextual reasoning across enterprise documentation.

Senior Software Engineer, Optum (UnitedHealth Group) **Jun 2021 – Jun 2024**

- Reduced time from ***500+ minutes to less than 3 minutes*** *by* refactoring the existing legacy system with high code quality standards.
- Optimized error report generation for the legacy system, reducing errors from ***25,000 to 20***, improving latency, and minimizing production costs through ***automated testing*** and ***CI/CD*** integration.
- Took initiatives to improve code quality & engineering principles by introducing design patterns, writing unit tests with at least 90% code coverage, creating code review guidelines, and mentoring juniors across teams with a growth mindset.
- Designed and implemented a distributed notification system for compliance notices as part of ***"The Paperless Initiative"***, using microservices architecture, processing millions of notifications and ***saving $500K annually.***
- Led a successful hackathon project on cardiovascular disease prediction, using ***data preprocessing, feature selection, and handling missing values***. Achieved 84.21% accuracy with ***Logistic Regression*** and 82.89% with ***Random Forest*** models.

Machine Learning Scientist Intern, Taiyo LLC **Dec 2019 – Apr 2020**

- Built an end-to-end ***ML pipeline for stock market prediction***, including ***time series forecasting, hyperparameter tuning across 10+ models***, and improving accuracy by **15%** on 50,000+ records.

SKILLS

Programming Language: Python, R, SQL, Java, C/C++, PostgreSQL, BigQuery, Javascript, Talend, Linux, Unix

Frameworks/Tools: Jenkins, Git, GCP, DevOps, Docker, Kubernetes, PySpark, Data Visualization, Databricks, Azure, AWS, GCP, ETL, Agile

AI/ML: PyTorch, Natural Language Processing (NLP), Neural Networks (including CNN, RNN, LSTM, GRU, FeedForward Networks), Deep Learning, Generative AI, Recommendation System, MLOps, Anomaly Detection, LLMs, Gen AI, Graph NN, Vertex AI, Prompt Engineering

Libraries/Software: Numpy, Matplotlib, Pandas, NLTK, SciKit-Learn, OpenCV, Grafana, Snowflake, StreamLit, FastAPI, Keras, TensorFlow

Core Computer Science:Data Structures, Algorithms, Distributed Computing, System Design, Transformer Models, Reinforcement learning

PUBLICATIONS

- **Bajpai A.,** D. Cavar, et al, Improving LLM Reasoning Through Ontology-driven Knowledge Graphs: A Comparative Study of Generating Ontologies for Medical RAGs, Midwest Speech and Language Days 2025, Indiana University Bloomington, April 2025.
- **Bajpai A.**, Garg M., Hindi Sentiment Analysis on Tweets, 2nd International Conference on Artificial Intelligence: Theory and Applications (AITA 2024). [SCOPUS][DBLP]

PROJECTS

Prostate Cancer Detection | *Computer Vision, Pytorch, CNN, U-Net, Image Segmentation* **Nov 2025 - Dec 2025**

- Built a **U-Net–based prostate MRI segmentation** pipeline using MONAI, comparing **zero-shot inference** with **fine-tuning** on medical imaging data; trained and evaluated on **IU BigRed200**, achieving a significant Dice score improvement **(0.415 to 0.774).**

AI for Early Kidney Disease Detection | *Snowflake, Streamlit* **Feb 2025 – May 2025**

- Led the project to perform ***data preprocessing, feature selection, data modeling,*** and ***data visualization***, building scalable pipelines, dashboards, and developed and **deployed ML models** for early kidney disease detection.

Rock Type Classification | *Machine learning, Ensemble Learning* **Sept 2024 – Oct 2024**

- Implemented and optimized **Softmax Regression, SVM (82.2%), and Random Forest (76.7%)** models for classifying rock types, leveraging features like texture and stripes. Developed and compared ensemble models **(Hard Voting: 81.1%, Soft Voting: 77.8%, Stacked: 78.9%)** against human classification accuracy to evaluate reliability and performance.